Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

Lütfullah Burak Kaya
Ozyegin University
burak.kaya.50711@ozu.edu.tr

Ismail Akturk
Ozyegin University
ismail.akturk@ozyegin.edu.tr

4 November 2026

gem5 TimingSimpleCPU · workload profile `test` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 8-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

Definitions (metric schema 9). Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	48 023 254 960	3 820 802 773	3 797 752 332	22 956 427	94 014	99.4	21.32	0.10	15.623	64.56	5.020	4.669	0.082	0.119
parest	37 262 401 281	130 109 038	16 148 624	110 058 554	3 901 860	12.4	0.22	0.59	9.895	67.70	1.416	1.473	0.225	0.627
deepsjeng	46 202 686 169	1 453 401 396	380 604 124	1 046 384 737	26 412 535	26.2	2.78	4.53	8.627	7.70	1.774	7.102	2.089	3.483
leela	26 063 891 644	808 702 134	26 356 444	749 189 404	33 156 286	3.3	0.36	5.75	5.161	27.19	0.591	0.422	1.011	0.368
namd	31 914 751 099	336 702 115	334 454 030	1 939 340	308 745	99.3	3.13	0.01	11.605	28.50	2.551	1.166	3.711	3.765
lbm	6 506 800 040	274 131 044	274 130 500	544	0	100.0	9.67	0.00	21.730	97.05	0.000	0.633	15.074	0.184
nab	6 691 483 051	152 456 167	124 535 928	24 348 560	3 571 679	81.7	7.90	0.73	4.074	7.87	0.156	0.129	0.628	0.732
xz	2 014 351 521	34 435 119	4 683 669	26 392 608	3 358 842	13.6	0.68	2.62	5.341	31.37	0.008	0.135	0.455	0.617
xz_4m_l1	2 001 462 938	30 569 523	3 959 703	23 548 312	3 061 508	13.0	0.59	2.35	6.788	41.37	0.009	0.167	0.994	0.605
xz_4m_l2	2 083 103 943	29 428 092	3 700 891	22 776 791	2 950 410	12.6	0.54	2.19	8.809	46.80	0.010	0.178	2.158	0.572
xz_4m_l3e	10 573 516 663	134 644 921	61 575 948	66 069 008	6 999 965	45.7	2.51	1.25	6.094	21.75	0.449	0.362	0.371	1.391
xz_4m_l4	4 011 967 585	68 464 242	20 213 033	44 411 548	3 839 661	29.5	2.12	2.21	6.791	37.27	0.230	0.751	0.552	0.305
xz_4m_l4e	12 115 711 352	137 799 962	63 036 496	67 532 844	7 230 622	45.7	2.28	1.11	5.967	19.86	0.573	0.377	0.399	1.663
xz_1m_l0	711 271 503	13 362 560	1 091 347	10 938 543	1 332 670	8.2	0.62	3.08	4.777	36.16	0.007	0.402	0.180	0.135
xz_1m_l1	719 848 327	12 739 426	973 549	10 458 508	1 307 369	7.6	0.58	2.91	5.395	42.00	0.007	0.444	0.308	0.188
xz_1m_l2	724 690 222	12 498 311	956 442	10 241 945	1 299 924	7.7	0.57	2.83	5.685	41.63	0.007	0.450	0.524	0.185
xz_1m_l3e	1 994 406 562	34 952 797	12 876 553	20 010 020	2 066 224	36.8	2.59	2.01	4.970	23.27	0.271	0.382	0.148	0.706
xz_1m_l4	1 076 057 785	23 230 841	5 156 845	16 486 476	1 587 520	22.2	2.01	3.06	5.046	32.77	0.090	0.779	0.249	0.180
xz_1m_l4e	2 198 787 666	35 410 889	13 089 895	20 222 586	2 098 408	37.0	2.43	1.84	5.016	21.50	0.367	0.391	0.154	0.842
imagick	117 733 556	2 444 490	2 450	2 437 082	4 958	0.1	0.02	4.14	4.168	86.22	7.194	58.765	0.298	0.307
blender	994 421 902	46 417 461	4 430 041	40 844 530	1 142 890	9.5	1.16	8.21	5.037	20.42	2.141	3.868	2.043	2.995
povray	2 240 986 446	168 867 726	2 121 674	155 851 654	10 894 398	1.3	0.27	13.91	8.697	47.73	6.932	19.449	1.477	1.013
omnetpp	12 426 267 350	842 780 888	5 233 033	791 025 406	46 522 449	0.6	0.09	12.73	11.572	28.59	0.053	0.721	1.230	2.192
xalancbmk	395 919 938	18 468 307	295 564	17 282 226	890 517	1.6	0.23	8.73	13.117	35.01	10.128	18.898	2.513	3.924
x264	475 535 613 470	19 842 562 937	13 777 288 367	5 322 257 634	743 016 936	69.4	6.24	2.24	1.858	32.44	0.158	0.861	0.937	1.233
gcc	16 026 850	1 021 294	52 739	952 288	16 267	5.2	0.84	11.88	5.733	38.80	0.234	1.328	0.443	0.672
mcf	24 426 982 047	234 360 896	91 410 910	142 147 076	802 910	39.0	1.58	1.16	24.643	78.48	3.545	6.915	2.675	5.246
perlbench_makerand	2 400 742 243	222 681 834	55	222 662 746	19 033	0.0	0.08	18.55	3.795	0.00	0.000	99.996	0.003	0.002
bwaves_1	17 288 358 363	145 483 800	145 451 376	26 481	5 943	100.0	3.55	0.00	26.018	37.38	5.467	0.021	3.308	4.973
bwaves_2	17 287 430 556	145 483 720	145 451 376	26 421	5 923	100.0	3.55	0.00	26.019	37.38	5.467	0.021	3.304	4.977
exchange2	57 667 143 431	1 874 880 445	1 304 771 747	345 493 349	224 615 349	69.6	5.03	1.20	1.342	2.01	1.430	2.700	3.470	2.262
fotonik3d	3 098 360 920	28 397 309	27 853 226	437 206	106 877	98.1	3.84	0.03	9.207	80.25	2.496	0.006	1.060	3.788
roms	28 955 766 090	203 007 943	182 777 748	12 238 510	7 991 685	90.0	4.79	0.08	11.833	67.05	2.452	0.885	0.302	2.456
wrf	84 396 891 740	420 514 378	109 794 207	285 402 322	25 317 849	26.1	0.91	0.68	6.920	55.78	6.233	4.759	0.462	1.292
cam4	33 466 803 539	585 237 550	168 288 974	404 426 599	12 521 977	28.8	2.11	2.42	8.840	60.99	1.527	2.519	0.613	1.469
Total	1 003 605 892 752	32 326 452 328	**21 110 519 840**	10 037 478 285	1 178 454 203	65.3	5.17	2.00	0.000	—	—	—	—	—